

MD RAKIBUL ISLAM RAIHAN

AI Engineer — LLM Infrastructure & Multimodal Systems (2027 New Grad)

Xi'an, China, Open to Relocation, Available July 2027

Nationality: Bangladesh, Sex: Male, DOB: 2002.12.06

Languages: Bengali (Native), English (Fluent), Chinese (Conversational)

Work Authorization: Requires China work visa (employer-sponsored Z Visa), Global: Sponsorship welcome

Email: raihanrakib.143@gmail.com Phone: +86 177 2236 2949 WeChat: imeddii Location: Xi'an, China

GitHub: github.com/theraihanrakib LinkedIn: linkedin.com/in/theraihanrakib Portfolio: theraihanrakibb.github.io/Online-Portfolio



SUMMARY

M.Eng. candidate in Software Engineering at NWPU (Top 1%, graduating July 2027), working where efficient generative AI meets the systems that actually deploy it. At **InfiX.ai** I cut LLM time-to-first-token **8.3x** on **4x A800** by building a three-tier KV-cache offload over Mooncake RDMA, then shipped PD-disaggregated **SGLang** serving that lifted stable concurrency from **192 to 256**. My first-authored IEEE TNNLS (Q1) paper, **NVFP4-DiT**, is a 4-bit audio-guided video diffusion transformer that cuts memory **4x** without losing quality, and my M.Eng. work covers multimodal audio-visual deepfake detection. I am looking for an AI-infrastructure or multimodal role where I can keep pairing modeling ideas with the CUDA/PyTorch systems work that gets them shipped.

EDUCATION

M.Eng. in Software Engineering, School of Software, Northwestern Polytechnical University (985/211) - Sep 2024 - Jul 2027

GPA: 88/100, Top 1%, Chinese Government Scholarship, President of International Student Union (ISU)

B.Eng. in Computer Science & Technology, School of Computer Science, Northwestern Polytechnical University (985/211) - Sep 2020 - Jul 2024

GPA: 85/100, Top 1%, Presidential Scholarship, Best Undergraduate Thesis Award

EXPERIENCE

AI Infrastructure Engineer Intern, InfiX.ai, Shenzhen, China, Apr 2026 - Jun 2026

- Cut TTFT 8.3x (55s → 6.7s) by building a 3-tier KV-cache offload (GPU HBM → CPU DRAM → SSD) over Mooncake RDMA on 4x A800, reaching 92% cache-hit for long-context models.
- Patched SGLang source across cache-management and memory-allocation modules to fix production cache failures and deploy hybrid-attention Qwen models.
- Built PD-disaggregated (prefill/decode) serving with RDMA KV sharing, raising stable concurrency 192 → 256 req and boosting GPU utilization via RadixAttention prefix-caching and continuous batching.
- Profiled CUDA/Triton inference with Nsight Systems; deployed Qwen on A800 and DeepSeek on H800 multi-node clusters, tuning tensor-parallel layouts and resolving 12+ production incidents (RDMA, etcd, storage).

Software Engineer Intern (AI Agent), Hong Kong Amram Group, Shenzhen, China, Jun 2025 - Sep 2025

- Developed production full-stack applications using React, REST APIs, and backend automation services.
- Built AI-agent workflow automation systems that improved operational efficiency by **25%**.

Electrical Software Engineer Intern, Shaanxi Longong Intelligent Technology, Xi'an, China, Feb 2025 - Apr 2025

- Developed PLC-based industrial control systems using CODESYS for robotics and industrial automation.
- Implemented Modbus, CAN, and Ethernet/IP communication protocols, improving motor-control performance by **15%**.

RESEARCH

Graduate Researcher, Northwestern Polytechnical University, Sep 2024 - Present

Advisor: Prof. Zhang Peng

- Developed **NVFP4-DiT**, a **4-bit audio-guided video diffusion transformer** achieving **4x memory reduction** while maintaining generation quality.
- Implemented low-precision quantization (**FP8/FP4/NVFP4**) for multimodal diffusion transformers; evaluated quality-efficiency trade-offs.
- Benchmarked NVFP4-DiT on **A800/H100 GPUs**; profiled end-to-end latency and memory with PyTorch Profiler and Nsight to guide low-precision kernel choices.
- Research on multimodal deepfake detection using visual, audio, and temporal consistency analysis.
- M.Eng. Thesis: "Detecting Deepfake Video by a Multimodal Audio-Visual Framework with Temporal Inconsistencies."

Undergraduate Researcher, Northwestern Polytechnical University, Jan 2023 - Jul 2024

Advisor: Dr. Yaxing Chen

- Designed and built a distributed confidential-query framework on Apache Spark + CKKS homomorphic encryption, achieving **95% privacy preservation** and **20% performance improvement** via optimized analytics pipelines.
- B.Eng. Thesis: "Design and Implementation of a Distributed Confidential Query Protocol for Apache Spark."

PROJECTS

AI Infra Portfolio, github.com/theraihanrakibb/ai-infra-portfolio

- 10 production-grade, Dockerized AI-infra projects: LLM gateway, model serving, GPU observability, training ops, MLOps registry, KV-cache simulator, benchmarking, quantization, tracing, RAG.

Distributed Training Launcher, github.com/theraihanrakibb/train-launcher

- CLI for multi-node PyTorch training with fault tolerance and automatic retry; cut manual ops overhead for distributed jobs.

NVFP4-DiT (Multimodal Generation), github.com/theraihanrakibb/NVFP4-DiT

- First-author IEEE TNNLS (Q1, under review) 4-bit audio-guided video diffusion transformer; 4x memory reduction while preserving generation quality on A800/H100.

Multimodal Deepfake Detection, github.com/theraihanrakibb/M.Eng-Thesis-Multimodal-Deepfake-Audio-Visual-Temporal-Framework

- M.Eng. thesis: audio-visual-temporal framework detecting deepfake video via visual, audio, and temporal inconsistency analysis.

TECHNICAL SKILLS

AI Infra & LLM Serving: SGLang, vLLM, TensorRT-LLM, KV-cache, PD-disaggregation, Mooncake RDMA

GPU & Low-Precision: CUDA, Triton, CUTLASS, FlashAttention, FP8/FP4/INT4 quantization, Nsight

Multimodal & Generative: PyTorch, Hugging Face Transformers, Diffusion (DiT), deepfake detection, OpenCV

Distributed & MLOps: DeepSpeed, Docker, Kubernetes, CI/CD, Prometheus, AWS, Linux

Languages & Fundamentals: Python (primary), C++, CUDA C++, Java, SQL, Bash, data structures & algorithms

PUBLICATIONS & HONORS

Publication: First Author, "NVFP4-DiT: Efficient 4-Bit Audio-Guided Video Diffusion Transformers for Low-Cost Video Generation" - IEEE Transactions on Neural Networks and Learning Systems (TNNLS, Q1), Under Review.

Honors: Chinese Government Scholarship (2024-2027); Presidential Scholarship (2020-2024); Best Undergraduate Thesis Award (2024); Outstanding Graduate (2024), NWPU; Hackathon Winner - NWPU AI Challenge (2023).

MD RAKIBUL ISLAM RAIHAN

AI 工程师 — 大模型基础设施与多模态系统 (2027 届应届)

现居: 中国·西安, 可接受异地, 2027 年 7 月到岗

国籍: 孟加拉国, 性别: 男, 出生日期: 2002.12.06

语言: 孟加拉语 (母语), 英语 (流利), 中文 (日常交流)

工作许可: 需办理中国工作签证 (雇主担保 Z 签证), 全球: 接受担保

邮箱: raihanrakib.143@gmail.com 电话: +86 177 2236 2949 微信: imeddii 地点: 西安, 中国

GitHub: github.com/theraihanrakibb

LinkedIn: linkedin.com/in/theraihanrakib

作品集: theraihanrakibb.github.io/Online-Portfolio



个人简历

西北工业大学软件工程硕士 (专业前 1%, 预计 2027 年 7 月毕业), 研究方向为高效生成式 AI 及其可部署系统。在 InfiX.ai 实习期间, 我在 4x A800 上基于 Mooncake RDMA 构建三级 KV-cache 卸载, 将大模型首字延迟降低 8.3 倍, 并通过 PD 分离的 SGLang 推理将稳定并发由 192 提升至 256。我的第一作者论文 NVFP4-DiT (IEEE TNNLS, 审稿中) 提出 4 比特音频引导视频扩散 Transformer, 在保持生成质量的同时显存降低 4 倍; 硕士课题同时涵盖多模态音视频深度伪造检测。希望加入 AI 基础设施或多模态团队, 持续把建模想法与 CUDA/PyTorch 系统实践结合落地。

教育背景

软件工程硕士, 软件学院, 西北工业大学 (985/211) - 2024年9月 - 2027年7月

GPA: 88/100, 前 1%, 中国政府奖学金, 国际学生联合会主席 (ISU)

计算机科学与技术学士, 计算机学院, 西北工业大学 (985/211) - 2020年9月 - 2024年7月

GPA: 85/100, 前 1%, 校长奖学金, 最佳本科毕业论文奖

工作经历

无界智索 (深圳) 科技 (InfiX.ai), AI 基础设施工程师实习, 深圳, 2026年4月 - 2026年6月

- 在 4x A800 上基于 Mooncake RDMA 构建三级 KV-cache 卸载 (GPU HBM -> CPU DRAM -> SSD), 将 TTFT 降低 8.3 倍 (55s -> 6.7s), 长上下文模型缓存命中率 92%。
- 修改 SGLang 源码的缓存管理与内存分配模块, 修复生产缓存故障并部署混合注意力 Qwen 模型。
- 构建 PD 分离 (prefill/decode) 推理架构, 基于 RDMA 共享 KV, 稳定并发由 192 -> 256 提升, 并通过 RadixAttention 前缀缓存与连续批处理提高 GPU 利用率。
- 使用 Nsight Systems 分析 CUDA/Triton 推理; 在 A800 部署 Qwen、在 H800 多节点集群部署 DeepSeek, 调优张量并行布局并解决 12+ 起生产故障 (RDMA, etcd, 存储)。

香港阿姆兰集团 (Amram Group), 软件工程师实习 (AI Agent), 深圳, 2025年6月 - 2025年9月

- 基于 React、REST API 与后端自动化服务开发生产级全栈应用。
- 开发 AI Agent 工作流自动化系统, 运营效率提升 25%。

陕西龙工智能科技 (Longong), 电气软件工程师实习, 西安, 2025年2月 - 2025年4月

- 基于 CODESYS 开发 PLC 工业控制系统, 用于机器人与工业自动化。
- 实现 Modbus、CAN 与 Ethernet/IP 通信协议, 电机控制性能提升 15%。

研究经历

研究生研究员, 西北工业大学, 2024年9月 - 至今

导师: 张蓬 教授

- 开发 NVFP4-DiT: 4 比特音频引导视频扩散 Transformer, 显存降低 4 倍且保持生成质量。
- 实现多模态扩散 Transformer 的低精度量化 (FP8/FP4/NVFP4), 评估质量-效率权衡。
- 在 A800/H100 上对 NVFP4-DiT 进行基准测试, 使用 PyTorch Profiler 与 Nsight 分析端到端延迟与显存, 指导低精度算子选型。
- 基于视觉、音频与时序一致性的多模态深度伪造检测研究。
- 硕士论文: 《检测深度伪造视频: 基于多模态音视频框架与时序不一致性》。

本科研究员, 西北工业大学, 2023年1月 - 2024年7月

导师: 陈亚星 博士

- 基于 Apache Spark 与 CKKS 同态加密设计并实现分布式机密查询框架, 通过优化分析管线实现 95% 隐私保护与 20% 性能提升。
- 本科论文: 《面向 Apache Spark 的分布式机密查询协议设计与实现》。

项目

AI 基础设施代码集, github.com/theraihanrakibb/ai-infra-portfolio

- 10 个生产级、Docker 化的 AI 基础设施项目: LLM 网关、模型服务、GPU 可观测性、训练运维、MLOps 注册表、KV-cache 模拟器、基准测试、量化、链路追踪、RAG。

分布式训练启动器, github.com/theraihanrakibb/train-launcher

- 用于启动与监控多节点 PyTorch 训练任务的命令行工具, 具备容错与自动重试; 降低分布式训练运维成本。

NVFP4-DiT (多模态生成), github.com/theraihanrakibb/NVFP4-DiT

- 第一作者 IEEE TNNLS (Q1, 审稿中) 4 比特音频引导视频扩散 Transformer; 在 A800/H100 上显存降低 4 倍且保持生成质量。

多模态深度伪造检测,

github.com/theraihanrakibb/M.Eng-Thesis-Multimodal-Deepfake-Audio-Visual-Temporal-Framework

- 硕士论文: 基于视觉、音频与时序不一致性分析的多模态深度伪造视频检测框架。

技术技能

大模型基础设施与服务: SGLang, vLLM, TensorRT-LLM, KV Cache, PD 分离, Mooncake RDMA

GPU 与低精度: CUDA, Triton, CUTLASS, FlashAttention, FP8/FP4/INT4 量化, Nsight

多模态与生成式: PyTorch, Hugging Face Transformers, Diffusion (DiT), 深度伪造检测, OpenCV

分布式与 MLOps: DeepSpeed, Docker, Kubernetes, CI/CD, Prometheus, AWS, Linux

语言与基础: Python (主语言), C++, CUDA C++, Java, SQL, Bash, 数据结构与算法

论文与荣誉

论文: 第一作者, 《NVFP4-DiT: 面向低成本视频生成的高效 4 比特音频引导视频扩散 Transformer》- IEEE TNNLS (Q1), 审稿中。

荣誉: 中国政府奖学金 (2024-2027); 校长奖学金 (2020-2024); 最佳本科毕业论文奖 (2024); 西北工业大学优秀毕业生 (2024); NWPU AI 挑战赛黑客松冠军 (2023)。